

A Data-Driven Optimization Framework for Profitable E-Waste Refurbishment in U.S. Recycling Systems

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Abstract—Electronic waste (e-waste) is one of the fastest-growing waste streams globally due to rapid technological advancement and device replacement cycles. Refurbishment and reuse of electronic devices represent an effective strategy for extending product life and reducing environmental impact. However, refurbishment decisions in small and medium-scale recycling operations are often based on intuition rather than systematic decision models, leading to inconsistent profitability and inefficient use of limited resources.

This paper proposes a decision optimization-driven information integration framework that combines machine learning-based profitability prediction with constrained binary optimization to support refurbishment selection decisions. A dataset of 51 refurbished electronic devices collected from operational refurbishment activities was constructed based on real operational refurbishment observations, including device condition indicators, repair costs, labor effort, and resale outcomes. Logistic Regression and Random Forest models were used to predict refurbishment profitability using stratified cross-validation. Model interpretability techniques were used to identify key economic drivers influencing profitability.

An optimization model was then created with respect to the selection of the appropriate set of devices to be refurbished, keeping in mind the budget and labor constraints. From the profitability prediction framework, the expected profits were obtained, which could be used as inputs to the optimization model and thus the refurbishment decisions could be made without the use of resources. Experimental results show that the Random Forest model achieved strong predictive performance ($F1 \approx 0.89$, $ROC-AUC \approx 0.97$). The optimization approach significantly improved total profit compared with heuristic selection strategies. The proposed framework demonstrates how operational refurbishment data can be transformed into structured decision-support intelligence for reverse logistics and circular economic operations.

Keywords: E-waste refurbishment; decision optimization; reverse logistics; machine learning; circular economy; resource-constrained selection.

I. INTRODUCTION

Electronic waste (e-waste) is one of the fastest-growing waste streams worldwide, driven by increasing device consumption and shorter product life cycles [1]. According to global reports, millions of tons of electronic devices are discarded every year, creating significant environmental and economic challenges.

Refurbishment and resale of electronic devices provide an important mechanism for extending device life cycles and reducing landfill waste [2]. By repairing and reusing used electronics, recycling organizations can recover economic value while supporting circular economy objectives.

Despite these benefits, refurbishment decisions in small and medium-scale recycling operations are often made using informal heuristics. Operators frequently prioritize newer devices or higher resale prices without systematically considering repair costs, labor requirements, and resource constraints. This can lead to unprofitable refurbishment decisions and inefficient resource allocation.

Previous research in reverse logistics has explored optimization approaches for product recovery and remanufacturing systems [3], [4]. However, limited research integrates predictive modeling and constrained optimization specifically for refurbishment decision-making in small-scale operational environments.

In addition, explainability and transparency in machine learning models have become important for operational decision systems, as decision makers must understand why certain recommendations are made [5].

To address these challenges, this study proposes a structured **information integration framework** that:

- Predicts refurbishment profitability using supervised machine learning
- Identifies key economic drivers using explainable modeling
- Selects optimal refurbishment portfolios under resource constraints

The proposed framework could be related to the themes of information reuse and integration, which are of central concern in the **IEEE International Conference on Information Reuse and Integration (IRI)**. In comparison with the previous research which focuses on single prediction and optimization tasks, this work combines interpretability of the machine learning model and the constrained optimization tasks and puts them into a common decision-support system for small-scale e-waste refurbishment businesses.

II. RELATED WORK

Reverse logistics and product recovery systems have been widely studied in supply chain management literature. Research by Fleischmann et al. [3] investigated the impact of product recovery on logistics network design, highlighting the complexity of recovery decision processes. Guide and Van Wassenhove [4] further examined challenges associated with profitability and uncertainty in closed-loop supply chains.

Recent studies have explored the role of machine learning in circular economic applications, including waste classification, demand prediction, and recycling logistics [7], [8]. However, the application of machine learning for predicting refurbishment profitability remains relatively underdeveloped.

Explainable artificial intelligence techniques such as SHAP and model feature importance have been applied to improve transparency in machine learning-based decision systems [5], [9]. These approaches help decision makers understand which variables influence predictive outcomes.

Optimization methods, particularly binary and integer programming models, have been widely used to solve resource allocation and decision optimization problems [10]. Such models allow decision makers to maximize profit or efficiency while respecting operational constraints such as budgets or labor availability.

This study differentiates itself by integrating:

- Machine learning–based profitability prediction
- Explainable model interpretation
- Resource-constrained optimization

into a unified refurbishment decision-support framework.

III. METHODOLOGY

Figure 1 illustrates the proposed data-driven refurbishment decision framework integrating profitability prediction and optimization-based device selection.

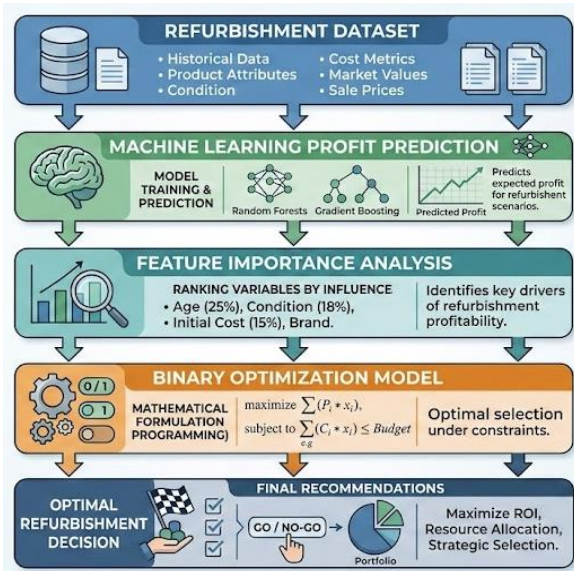


FIG. 1. DATA-DRIVEN REFURBISHMENT DECISION FRAMEWORK

A. Dataset

The dataset consists of records of refurbishment history of 51 operational electronic devices from actual refurbishment practices, such as smartphones and laptops. Based on the output of the realized profit, the devices were classified as profitable and non-profitable ones to be used for the training and testing of the model. This classification is useful for better understanding model results and for assessing the success of profit prediction.

The dataset includes the following categories of variables used for profitability prediction and decision optimization.

Device attributes: device age, storage capacity, and battery health percentage

Condition indicators: screen damage, body damage, water damage, and functional status prior to repair

Cost variables: purchase cost, parts cost, labor hours, labor cost, and total refurbishment cost

Outcome variables: estimated resale price and realized profit

For each device i , realized profit is defined as:

$$\text{Profit}_i = \text{ResalePrice}_i - \text{TotalRefurbishmentCost}_i$$

A binary profitability label is then assigned as:

$$Y_i = \begin{cases} 1, & \text{if } \text{Profit}_i > 0 \\ 0, & \text{otherwise} \end{cases}$$

This label indicates whether refurbishing device i resulted in a profitable outcome.

B. Profitability Prediction

Each device instance is represented by a feature vector:

$$X_i = (\text{Age}, \text{BatteryHealth}, \text{ScreenDamage}, \text{BodyDamage}, \text{WaterDamage}, \text{Functional Status}, \text{PurchaseCost}, \text{PartsCost}, \text{LaborHours}, \text{LaborCost})$$

Two supervised learning models were evaluated:

- Logistic Regression (baseline linear classifier)
- Random Forest (ensemble decision-tree model)

Logistic Regression provides a baseline linear classification model, while Random Forest is an ensemble method capable of capturing nonlinear relationships between variables.

Stratified five-fold cross-validation was used to maintain the benefit class distribution of profitable/caseless of a model to evaluate the performance. To minimize variance in the performance of an averaged results over all folds were used, thus eliminating unreliable performance estimation models. The following metrics were used:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

Feature importance was calculated using random forest impurity based feature importance scores, to identify the most influential drivers which effect the profitability of refurbishment. The final systems subsequently yield rankings that better inform model interpretability by highlighting the importance of models for predicting profitability and refitting the vessel.

C. Decision Optimization Model

In practical deployment, refurbishment decisions must be made before actual resale outcomes are known. Therefore, the optimization model uses expected profit estimates generated by the machine learning prediction framework, while realized profits are used only for retrospective evaluation.

Maximize:

$$\max \sum_{i=1}^n Profit_i x_i$$

Subject to:

$$\begin{aligned} \sum_{i=1}^n Cost_i x_i &\leq B \\ \sum_{i=1}^n LaborHour_i x_i &\leq H \\ x_i &\in \{0,1\} \end{aligned}$$

where:

B represents the available refurbishment budget, H represents available labor capacity, and x_i indicates whether device i is selected for refurbishment.

IV. Experimental Results

A. Predictive Model Performance

The performance of the evaluated machine learning models is summarized in Table I.

TABLE I
Predictive Performance of Classification Models

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.80	0.88	0.83	0.85	0.89
Random Forest	0.84	0.86	0.94	0.89	0.97

The Random Forest model achieved the highest performance, particularly in recall and ROC-AUC, indicating strong capability in identifying profitable refurbishment opportunities.

Analysis of feature importance showed that **economic factors**, including purchase cost, labor cost, and total refurbishment cost, played the largest role in predicting profitability.

B. Optimization Under Resource Constraints

To evaluate the operational benefit of the proposed framework, a constrained refurbishment scenario was simulated with the following limits:

- Budget: \$1500
- Labor **availability**: 10 hours

Two strategies were compared:

1. **Resale-value heuristic**: selecting devices with the highest resale price
2. **Optimization model**: selecting devices that maximize total profit

The results are shown in Table II.

TABLE II
RESOURCE-CONSTRAINED REFURBISHMENT SELECTION

Method	Total Profit
High-Resale Heuristic	\$235
Optimization Model	\$1033

The optimization model significantly improved profit compared to the heuristic strategy, demonstrating the effectiveness of integrating predictive modeling with decision optimization.

V. Discussion

The results indicate that refurbishment profitability is primarily driven by economic variables rather than device brand or nominal resale value. Cost-related features such as purchase cost, parts cost, and labor requirements dominate decision outcomes.

By integrating machine learning prediction with constrained optimization, the proposed framework enables structured decision reuse across refurbishment operations. This approach transforms operational refurbishment data into actionable decision-support intelligence, consistent with reverse logistics research and circular economic objectives.

VI. Conclusion

This study introduced a **decision optimization-driven information integration framework** for electronic waste refurbishment operations. By combining predictive machine learning models, explainability mechanisms, and constrained optimization techniques, the proposed approach improves profitability and resource utilization in refurbishment decision-making.

Experimental results demonstrate strong predictive performance and show that optimization-based selection significantly outperforms resale-driven heuristics under limited resources.

A drawback of this study is that only 51 of the refurbishment records are available for this data set. Although the results have shown that the proposed framework is feasible, there is a need for more extensive data to be collected to see if this framework holds true across various environments where refurbishment may take place.

Future research will be extended to include more data and include further market variables including resale price volatility and uncertainty of demand.

VII. REFERENCES

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